

The hole-Nyquist defect

Test polynomials for $d_i \geq 1$, and why the off-diagonal mass is not a Fourier or Bernstein theorem

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Abstract

After r407 ($P1 \iff \lambda_2(C) \geq 1$) and r408 ($d_{\min} \geq 1$ is the necessary half, the exception mode is hole-Nyquist), this note attacks both halves. Half N: the reproducing-kernel test-polynomial bound with the Y -Lagrange interpolant (degree $|Y| - 1 < n$) does *not* prove $d_i \geq 1$ ($b_{\max} = 0.071$, zero holes cleared). Half S: C is not banded in the hole-DFT (off-fraction 0.968); $\|T_0 v_{\text{Nyq}}\| < \|T_0\|$; the defect is born at $k = 37$ along the θ -prefix and stays unique. No RH claim.

Status. Lemmas 1–4 are proved over $\mathbb{Q}$. The test-polynomial SATZ-candidate, the Fourier-block candidate, and the Bernstein candidate are **refuted**. $d_i \geq 1$ on the source family is a census. The sequential birth is a named reduction, not a theorem. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The object

$C = BB^T$ is the μ -only dual CD Gram on Y (r407),

$$d_i = C_{ii} = u^\vee(y_i) K_n^\mu(y_i, y_i).$$

Rayleigh gives $d_{\min} < 1 \Rightarrow \lambda_{\min} < 1$. r408: source $d_{\min} \in [1.097, 1.656]$; scramble 0.37; permute 0.15 (same Y).

2 Half N: the test polynomial

Lemma 1 (Reproducing kernel). *For $p \in \mathcal{P}_n$, $K_n^\mu(y, y) \geq |p(y)|^2 / \|p\|_{L^2(\mu)}^2$. On three unit atoms, $n = 1$, $K = 1/h_0 = 1/3$.*

Lemma 2 (Degree obstruction). *If $\deg p \geq n$ the inequality may fail: $p(t) = t^2$ on $\{-1, 0, 1\}$, $n = 1$, gives $|p(1)|^2 / \|p\|^2 = 1/2 > 1/3 = K$.*

Lemma 3 ($C_{ii} = u^\vee K$). *$C = BB^T$ with $B_{ki} = \sqrt{u^\vee(y_k)} P_i(y_k)$ implies $C_{ii} = u^\vee(y_i) \sum_{j < n} P_j(y_i)^2$.*

Lemma 4 (Rayleigh). $\lambda_{\min}(C) \leq \min_i C_{ii}$.

The candidate p_{y_i} is the Y -Lagrange basis ℓ_i (zeros on $Y \setminus \{y_i\}$, $\ell_i(y_i) = 1$). On FRAME-A, $\deg \ell_i = 103 < n = 181$, so Lemma 1 applies, and

$$d_i \geq b_i := u^\vee(y_i) / \sum_{x \in X} u^\vee(x) \ell_i(x)^2.$$

Proposition 1 (Lagrange does not close). *On $kz = 9$, $b_{\max} = 0.071$, $\#\{b_i \geq 1\} = 0/104$, $\min(d_i/b_i) = 94$ (no RK violation). Local Lagrange is worse. The Chebyshev-CD test of degree $n - 1$ clears 81/104 holes and leaves $b_{\min} = 4.9 \cdot 10^{-4}$.*

So Half N is **not** a SATZ by this family of explicit p . $d_i \geq 1$ remains a source census (core-42, $d_{\min} \in [1.097, 1.656]$).

Permute (same Y) drops $d_{\min}$ to 0.148 by a *Christoffel* collapse: $K_n^\mu(y, y) = 0.148$ at a hole with $u^\vee = 1$. The test-poly bounds stay tiny on scramble and permute — they do not exhibit the break. Live and dead χ_3 keep $d_{\min} > 1$.

3 Half S: one Nyquist mode

Proposition 2 (Fourier is not a block). *The hole-DFT of C has off-diagonal energy fraction 0.968; an 8-diagonal band holds only 0.18 of the energy. The Nyquist bin is not a spectral atom.*

Proposition 3 (Bernstein is not tight). $\|T_0 v_{\text{Nyq}}\| = 0.656 < \|T_0\| = 1.080$ (*r409 graph operator*). *The defect eigenvector has hole-Nyquist overlap 0.386, not 1. A Markov estimate of the required slope over the minimal x -gap is 0.14 of the n^2 bound — not critical.*

Proposition 4 (Sequential birth). *Rebuilding C along the θ -prefix of Y : $k = 32$ has $n_C = 0$, $C_{\min} = 1.00014$; $k = 37$ has $n_C = 1$, $C_{\min} = 0.99991$; $k = 104$ has $n_C = 1$, $\lambda_2 = 1.00018$. The defect is born once and does not split. The flip index depends on order (a random prefix flips later). This reduces Half S to a one-path census, not a Cauchy theorem (rebuilt C is not a principal submatrix).*

Remark 1. P1 remains equivalent to $\lambda_2(C) \geq 1$. Neither half proves the right-hand side. Unconditional SATZ is Lemmas 1–4. 1010 at $S = 20$ has $\deg \ell > |Y|$ vs. n and violates RK ($d < b$), as predicted by Lemma 2.

Kill table (machine)

mutant	effect	what it kills
Y-Lagrange as SATZ	$b_{\max} = 0.071$	Half N proof
Chebyshev-CD $m = n$	81/104	remaining holes
scramble	$d_{\min} = 0.37$	Half N on that world
Λ -permute	$K \rightarrow 0.148$	same Y , weights
1010 $S = 20$	$\deg > n, d < b$	RK hypothesis
flip +10	$n_C = 8$	density
depth $n + 1$	$C_{\min} \rightarrow 1$	razor